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Sulphates:

The sulphates of the alkaline earth metals are all white solids and stable to heat. $BeSO_4$, and $MgSO_4$ are readily soluble in water; the solubility decreases from $CaSO_4$ to $BaSO_4$. The greater hydration enthalpies of Be^{2+} and Mg^{2+} ions overcome the lattice enthalpy factor and therefore their sulphates are soluble in water.

Nitrates:

The nitrates are made by dissolution of the carbonates in dilute nitric acid. Magnesium nitrate crystallises with six molecules of water, whereas barium nitrate crystallises as the anhydrous salt. This again shows a decreasing tendency to form hydrates with increasing size. All of them decompose on heating to give the oxide.

5.6.1 Important compounds of calcium

Quick lime, CaO

Preparation

It is produced on a commercial scale by heating limestone in a lime kiln in the temperature range 1070-1270K.

$$CaCO_3 \rightleftharpoons CaO + CO_2$$

The reaction being reversible, carbon dioxide is removed as soon as it is produced to enable the reaction to proceed to completion.

Properties

Calcium oxide is a white amorphous solid.

It has a melting point of 2870 K.

It absorbs moisture and carbon dioxide on exposure to atmosphere.

$$CaO + H_2O \rightarrow Ca(OH)_2$$

$$CaO + CO_2 \rightarrow CaCO_3$$

The addition of limited amount of water breaks the lump of lime. This process is called slaking of lime and the product is slaked lime.

$$CaO + H_2O \rightarrow Ca(OH)_2$$

Quick lime mixed with soda gives solid soda lime. It combines with acidic oxides such as SiO_2 and P_4O_{10} to form $CaSiO_3$ and $Ca_3(PO_4)_2$, respectively.

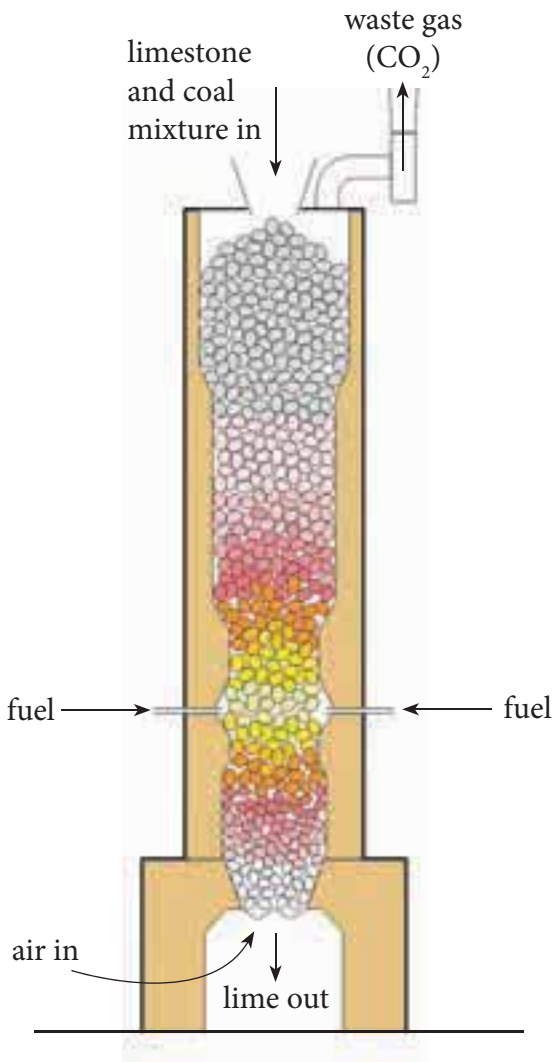


Figure 5.10 Preparation of Quick Lime